On the observed scaling effects in language model predictions of reading time

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The time taken to read a word is highly correlated with its statistical properties. More frequent words are read faster than less frequent words (Rayner, 1977), and words with a higher statistical probability in context are read faster than lower-probability words (McDonald and Shillcock, 2003). While for many years the evidence suggested that more powerful statistical models of language provided a better fit to the data (Goodkind and Bicknell, 2018; Wilcox et al., 2023), recent work has shown that modern neural language models may become 'too good' at next word prediction and begin to show a worse fit to reading time (Kuribayashi et al., 2021; Oh et al., 2022); in particular, they appear to under-estimate reading time for low-frequency words (Oh et al., 2024). Additionally, despite the fact that the predictions of language models ostensibly capture the statistical probability of words, work has shown that lexical frequency explains additional variance in reading time (Shain, 2024); and this is true to a greater extent for models that have become 'too good' at next word prediction (Oh and Schuler, 2025). How should we interpret these findings?

We argue that a crucial piece of the puzzle may be the finding that the stage of training at which language models show the best fit to reading time (Oh and Schuler, 2023) appears to correspond to the stage that elsewhere has been been shown to be when their predictions most closely correlate with lower-order n-grams (Michaelov et al., 2025). If so, this suggests that rather than thinking of language models as improving in fit as they improve at next-word prediction until they become 'too good', it may instead be more useful to think of models as becoming more and then eventually less n-gram-like in their predictions. We test whether this hypothesis holds in the data.

Method We take the item-level mean of the first fixation duration, first pass duration, go-past duration, and total duration of all words in the Provo Corpus (Luke and Christianson, 2018). We calculate surprisal for all of these words using each of the main Pythia models (Biderman et al., 2023) at 20 stages over the course of training. We also calculate the unigram, bigram, and trigram surprisal of each word in the Pythia training corpus (Gao et al., 2020; Liu et al., 2024). We then calculate the correlations between Pythia surprisals and the reading time and n-gram metrics.

Results First, we compare trajectory of the correlation between Pythia surprisal and reading time to that of the correlation between Pythia surprisal and n-gram surprisal, which we present in Figure 1A. As can be observed, the degree to which first fixation duration is correlated with Pythia surprisal shows a highly similar time course to trigram surprisal, while the time course for the other metrics of reading time more closely resemble those of unigram and bigram surprisal.

To further compare the synchronicity between fit to these variables, we calculate the correlation between these correlations (Figure 1B). We observe a clear positive Pearson correlation r between Pythia surprisal's correlation with n-gram surprisal and its correlation with first fixation duration (unigram: 0.57; bigram: 0.95; trigram: 0.99), first pass duration (unigram: 0.80; bigram: 0.99; trigram: 0.93), go-past duration (unigram: 0.87; bigram: 0.97; trigram: 0.85), and total duration (unigram: 0.66; bigram: 0.98; trigram: 0.98).

Discussion Our results suggest that when a given language model is at its most n-gram-like, it is also best at predicting reading time, with the most consistent pattern observed for bigram-likeness (all rs$\geq$0.95). This provides a unifying perspective on the aforementioned and apparently disparate effects reported in previous work. A possible implication of our results is that the neurocognitive systems engaged in eye movement during naturalistic reading may be sensitive to surface-level n-gram-like statistics rather than the complex language statistics that are learned by more complex neural language models trained on many billions (or even trillions) of tokens. In future work, we plan to test this hypothesis with a greater range of datasets, languages, and language models.

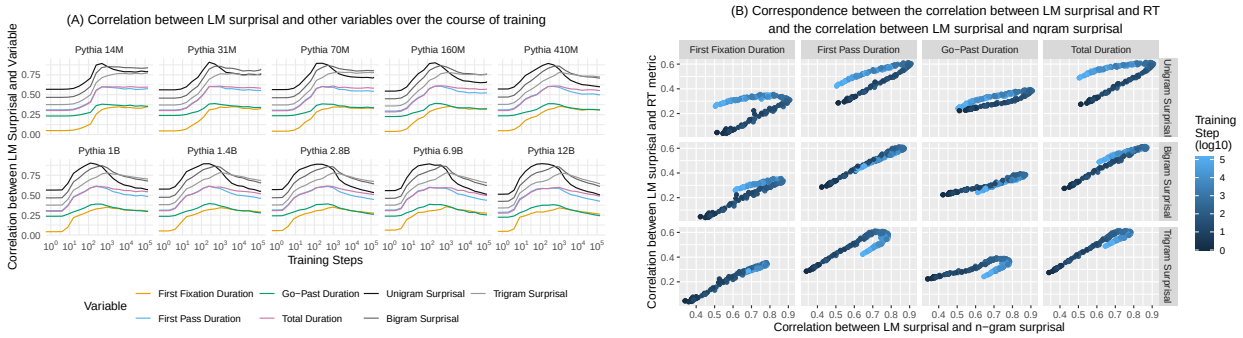

Figure 1: How language model surprisal correlates with n-gram surprisal and reading time.

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